Kunal 3:13:19

Network layer.

→ Here we work with Router.

→ Logical Addressing. Provide IP Add to Identify devices on a Network.

192.168.2.30

Network Address — Device address.

→ Routing : Determine best path for data packets to travel from source to dest.

→ Packet forwarding : Transfer packets through router.

Control plane : This is used to build the router tables and every router like a graph

Router → Nodes.

Links → Edges.

Further no need to make notes. watch vid of Kunal. ☺

Two types of Routing

- **Static Router**
  → Manually add all Routes.
- **Dynamic Routing.**
  → Evolve accordingly when it need.

Internet protocol (IP)

Reserved

Network Id | Host

0 ------- . [ ] . [ ] . [ ]

8 bit 8 bit 8 bit

0 → 00000000
00000001
0000010
127 → 01111111

this two are not used

① No of Network in CA = $2^7 = (128-2)$
= 126 used.

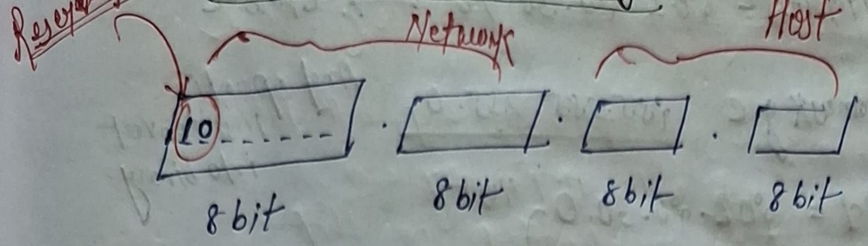
No of Host possible in every Network = $2^{24} - 2$

Class A Default Mask = 255.0.0.0

64.0.0.8

And
01000000.0.0.00001000
11111111.0.0.00000000
64.0.0.0

Class B IP Addressing.



128 10000000
10000000
10000000
⋮
191 10111111

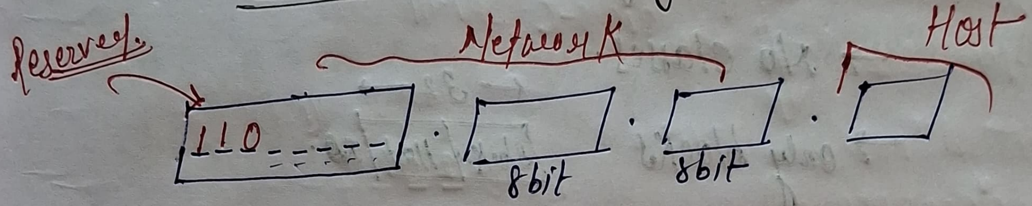
Range - 128-191.

No of Add in B = $2^{30} \approx$ 1 bilon.

No of Network - $2^{14} \approx 16384$.

No of Host in each Network = $2^{16} \approx 65536$.

Class-C IP Addressing



192 11000000
11000001
⋮
223 11011111

Range = 192-223

No of IP Add = 2^{29}

No of Network = 2^{21}

" Host = 254

Class-D IP Add.

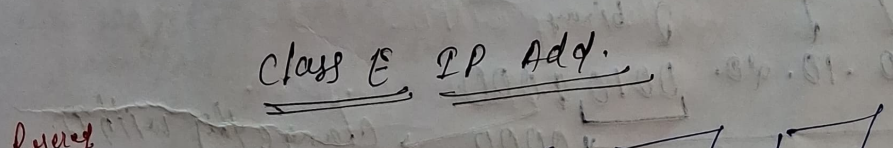
224 11100000
11100001
239 11101111

Range - 224-239.

IP Add = 2^{28}

× No of Net × NO HOST also.

Class E IP Add.



Range - 240-255

Q. IP Address = 201.20.30.40 → class-C

The default mask of C 255.255.255.0 — Any operation first convert into binary.

So Network Id = 201.20.30.0 Host

4th Host Id = 201.20.30.4

Last Host Id = 201.20.30.255 we never give last broadcast

↳ 201.20.30.254

Classless Addressing (CIDR).

→ No classes.

→ only blocks ← 32 → | Block of Host | 32.

→ Notation x.y.z.w/n → mask

→ 200.10.20.48/28 → Block   32 − 28 = 4 → Host

28 no of 1s.   No of Host = 2^4

11111111.11111111.11111111.11110000

8 bit   8 bit   8 bit   total 28 1s are there

$\boxed{255.255.255.240}$ → Mask of Net

Now, 200.10.20.48

↓ ↓ 2 binary

200.10.20.0010,1000

← remain same → 0000 → change this with 0.

$\boxed{200.10.20.32}$ Network Id.

Rule:- go Chatgpt, last time work!

Subnetting in classful Addressing.

Dividing the big network into smaller network.
Maintenance can easier.

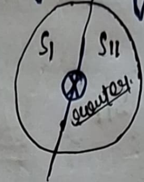
taking example of class-C (200.10.20.0)

0 – 255] 254 used.

In Subnetting we not change any bit in (Networking bit)

Subnet I

200.10.20.00000000
(0) Fix
0 0000001
0 0000010
⋮
(127) 0 1111111



Subnet II

200.10.20.1 _ _ _ _ _ _ _
128 Fix 1 0000001
1 0000010
(255) 1 1111111

200.10.20.0 to 200.10.20.127 | 200.10.20.128 to 200.10.10.255.

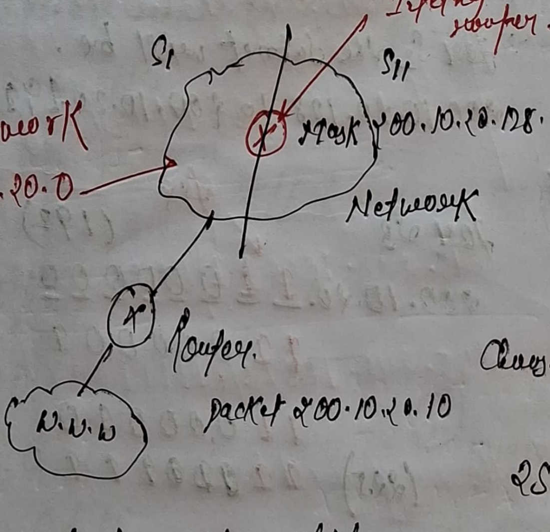
Subnet Id — broadcast Ad

Total Host Add for S_1 - 126 . S_{11} - 126

Default Mask for class-C 255.255.255.0

EX-01

Whole Network 200.10.20.0



(i) if any packet will come to that outer Router
let packet 200.10.20.10.
then to know which home/Network that pack should chose then it will take
Class-C Default Mask

255.255.255.0.
(And)
200.10.20.10

200.10.20.0

Now easily Home/Network determine where that data packet should go.

(ii) Now we got the home Now which room/Application this data will go. By Subnetting.
→ Inside the Internet one more router will have to placed.
→ Now Internal router we have to give

Subnet Mask : S_1 200.10.20.0|0000000, 1 bit fix
10000000 (128)
200.10.20.128

∵ To know which Subnet the packet come by (AND) operation packet & Subnet Mask.

Variable length Subnet Masking (VLSM)

⇓

We have to convert larger network into small network but not equaly one should big and one can smaller in that case we use Variable length Subnet Masking.

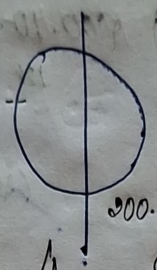
In that

0] (254)
255] We can use.

S_1 S_{11}

200.10.20.0 ----

fix

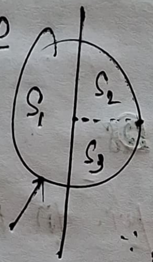


200.10.20.0

class-C IP Add.

for S_1 :- fix

200.10.20.00000000
0 0000001
0 0000010
⋮
(127) 0 1111111



for S_1 subnet the range will be.
200.10.20.0 to 200.10.20.127.

fix (128)

200.10.20. 1 0 0 0 0 0 0 0
1 0 0 0 0 0 0 1
1 0 0 0 0 0 1 0
⋮
(192) 1 0 1 1 1 1 1 1

∴ for S_2 the subnet will be.
200.10.20.128 to 200.10.20.191.

for S_3 fix (192)

200.10.20. 1 1 0 0 0 0 0 0
1 1 0 0 0 0 0 1
⋮
1 1 0 0 0 0 1 0
(225) 1 1 0 0 0 1 1 1

So the rang will be
200.10.20.192 to 200.10.20.225.

Subnetting in classless Interdomain Routing

"(Subnetting in CIDR)"

195.10.20.128/26 → 26 bit is Network bit.

Network bit

195.10.20.10000000

8 bit 8 bit 8 bit

FIX

195.10.20.10000000

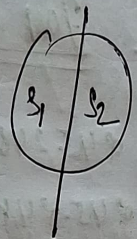
Network bit

195.10.20.10000000

10000001

⋮

10011111



Network. FIX

195.10.20.10100000

10100001

⋮

10111111

195.10.20.128/27 to

195.10.20.159/27

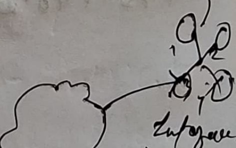
195.10.20.160/27 to

195.10.20.191/27

lec-52

Q.1. Classless interdomain routing (CIDR) receive a packet with address 131.23.151.76. The router's routing table has following entities.

Prefix	Output Interface
131.16.0.0/12 (continuous one / other host)	3
131.28.0.0/14	5
131.19.0.0/16	2
131.22.0.0/15	1



(1) 131.23.151.76.

11111111.11110000.00000000.00000000

convert → in binary number then AND operation.

after, again convert to normal no. (deci) then match that no with the given prefix number, if match then that select.

2^5 2^4 2^3 2^2 2^1 2^0

16 8 6 4 2 1

Variable length Subnet Masking in (CIDR)

245.248.128.0/20 ✓ Network bit.

Network bit

245.248.1000|0000.00000000

8bit 8bit 8bit 8bit

S1

245.248.10000 0000.

00000000

245.248.10000 001.00000001

245.248.10000 010.00000010.

245.248.10000 111.11111111

So the rang will be.

245.248.128.0/21 to

245.248.135.255/21

For S2:

Fix

245.248.10001|000:00000000

245.248.10001000.00000001

:

245.248.10001011.11111111.

139 255

the range of S2 will be

245.248.136.0 to 245.248.139.225.

For S3 Subnet Masking.

Fix.

245.248.10001 1|000.00000000

245.248.100011|11.11111111.

So the rang will be

245.248.148.0 to 245.248.143.255.

IPv4 Header format

⇓

The IPv4 header is 20 to 60 bytes long and contains information needed to deliver a packet from source to destination.

→ TTL. Protocol, and checksum which help in routing and delivering the packet correctly across network.

→ connectionless, datagram service.

Header → payload.

VER 4	HLEN 4	Type of service (DSCP) 8	Total length 16.
Identification bits 16	flag 3	fragment offset 13	
Time to LIVE TTL 8	Protocol 8	Header checksum 16.	
Source IP Address 32 bits.			
destination IP Add 32 bits.			
→ options & padding.			

fragmentation IPv4 datagram

A datagram of 3000B (20B of IP Header + 2980B IP payload) reached at Router and must be forward to link with MTU of 500 B. How many fragmentation will be generated and also write MF, offset, Total value for all.

2980B/480 = 7 fragment form.

Router → 500B (capacity)

3000 → 20 header, 2980 payload

500B → 20 + 480

P7 P6 P5 P4 P3 P2 P1

100+20, " " " " 480+20, 480+20

2880

Refer last page

(ii) MF (More fragment) → if a packet is let, P_1 then is there any other packet coming after? If Yes (1), If No (0)

P_7	P_6	P_5	P_4	P_3	P_2	P_1
(0)	(1)	(1)	(1)	(1)	(1)	(1)

after that No packet having data so, P_7 (0).

(iii) Offset: How much bite of data is in upcoming packet, if there is a data then div by (8).

P_7	P_6	P_5	P_4	P_3	P_2	P_1
400+20	480+20	480+20	480+20	480+20	480+20	480+20
			(480×3)/8	(480+480)/8	480/8	0
360	300	240	=180	=128	=60	(after P_1 no packet)

Need of IPv6 Protocol

→ Limitation in IPv4 Addresses. (2^{32})

→ Real time data transmission.

→ Authentication

→ Encryption enabled.

→ fast processing at routers.

IPv6 Header.

⇓

It's introduced to overcome the limitations of IPv4, especially the shortage of IP address, provides a much larger 128 bits addressing space.

(i) Large Add Space → 128-bit addressing.

(ii) Solves Address Exhaustion: Provide enough IP addresses for computer, smartphone, IOT devices.

(iii) Simplified Header: IPv6 has simpler header than IPv4

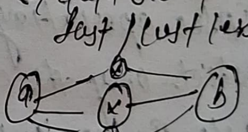
(iv) Better Security (v) Improved Router Efficiency.

the more prior data go first

for real time data processing.

VERSION	Priority (8)	Flow label (20)	
Payload length (16)		Next (8) Header	Hop (8) Limit
Source Address (128)			
Destination Address (128)			

→ data sharing fast / lost less.



can add extension also.

Base Header = 40 Bytes (320 bits) Fixed.

Extension Header.

1. Routing Header
2. Hop by Hop.
3. Fragment Header
4. Authentication Header
5. Destination Header.
6. Encapsulating Security Payload (ESP)

IP Security protocol: (IPsec)

→ Network layer protocol. (both IPv6 & IPv4)

→ It is a collection of protocol.

- Encapsulating Security Payload (ESP).
- Authentication Header (AH)
- Internet Key exchange.

Uses of IP Security protocol

- confidentiality.
- Authentication
- Replay message Attack

They work in (Transport Mode & Tunnel Mode)

$\underbrace{\text{Transport Mode \& Tunnel Mode}}_{\text{Mode of operation.}}$

Transport Mode & Tunnel Mode. L-60.

Transport Model is a IPSec operating mode in which only the data (playload) of the IP packet is encrypted / authenticated, while the original IP header remain unchanged.

Host to Host communication.

Tunnel Mode is an IPsec operating mode in which the entire original IP packet (header + data) is encrypted and encapsulated inside a new IP packet with new IP header.

Gateway to Gateway Comm

Routing Protocols.

It helps to find the best route for forwarding data packets from source to destination.

Intra domain Routing

An intra-domain routing protocol is used to exchange routing information within same Organisation.

- Distance Vector (RIP)
- Link State (OSPF)

Inter domain Routing.

It used to exchange the routes between different autonomous system

- Path Vector (BGP).